



**Department of Information Technology**  
**Academic Year 2024 – 2025(Even Semester)**

**Degree, Semester & Branch: B.Tech-VI & IT**

**Course Code & Title: CCS356 & Object-Oriented Software Engineering**

**Name of the Faculty member (s): Mrs.M. Rethinakumari**

### **Innovative Practice Description**

- **Unit / Topic: Unit V / Deployment Tools**

**Course Outcome:** The student will be able to understand various deployment tools used in modern software engineering. They can also analyse the features and functionalities of tools like Jenkins, Docker, and Kubernetes. They can apply appropriate deployment tools to automate software delivery processes and demonstrate the ability to configure and manage deployment environments effectively.

**Activity Chosen:** Think Pair Share

- **Justification:**

- The Think-Pair-Share strategy helps students process new concepts, share perspectives, and develop a clearer understanding through peer discussion.
- **Deployment tools** such as Jenkins, Docker, and Kubernetes are essential in modern software development, enabling automation, scalability, and efficient deployment.
- Through this activity, students gain exposure to real-world tools and practices used in DevOps environments.
- Collaborative learning enhances students' ability to analyse, evaluate, and apply deployment tools in various software engineering contexts.

**Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- The instructor delivered a 30-minute session explaining the purpose, features, and uses of deployment tools like Jenkins (automation server), Docker (containerization), and Kubernetes (orchestration).
- The key questions was provided (e.g., "What are the roles of Jenkins, Docker, and Kubernetes in software deployment?").
- **Think:** Students independently reflected on each tool's purpose and workflow.
- **Pair:** Students discussed their thoughts with a partner, clarified doubts, and explored practical scenarios.
- **Share:** Each pair shared their understanding with the class. The instructor moderated, addressed misconceptions, and emphasized best practices. The practice attached as the figures 1 and 2.



• **CO – PO / PSO mapping:**

| CO  | PO1 | PO3 | PO5 | PO8 | PO9 | PSO1 | PSO2 |
|-----|-----|-----|-----|-----|-----|------|------|
| CO4 | 3   | 2   | 3   | 3   | 1   | 3    | 1    |

(1 – Low

2 – Moderate

3 – High)

• **PO / PSO mapped:**

| Innovative practice                  | PO1                                                                                                                                                  | PO3                                                                                                                                                  | PO5                                                                                                                                                  | PO8                                                                                                                                                      | PO9                                                                       | PSO1                                                                                                                                           | PSO2                                                                                                                                                      |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Justification for correlation</b> | Apply engineering principles to efficiently plan and manage software projects. Ensure effective utilization of resources and adherence to timelines. | Apply engineering principles to efficiently plan and manage software projects. Ensure effective utilization of resources and adherence to timelines. | Apply engineering principles to efficiently plan and manage software projects. Ensure effective utilization of resources and adherence to timelines. | Follow ethical principles during project management, ensuring fairness in resource allocation and decision-making. Promote accountability and integrity. | Collaborate effectively in teams to manage and execute software projects. | Use cloud platforms and tools for efficient project execution and delivery. Integrate DevOps practices for seamless operation and scalability. | Deliver industry-ready software products by implementing systematic project management strategies. Ensure quality assurance through continuous delivery.. |

• **Images / Screenshot of the practice:**

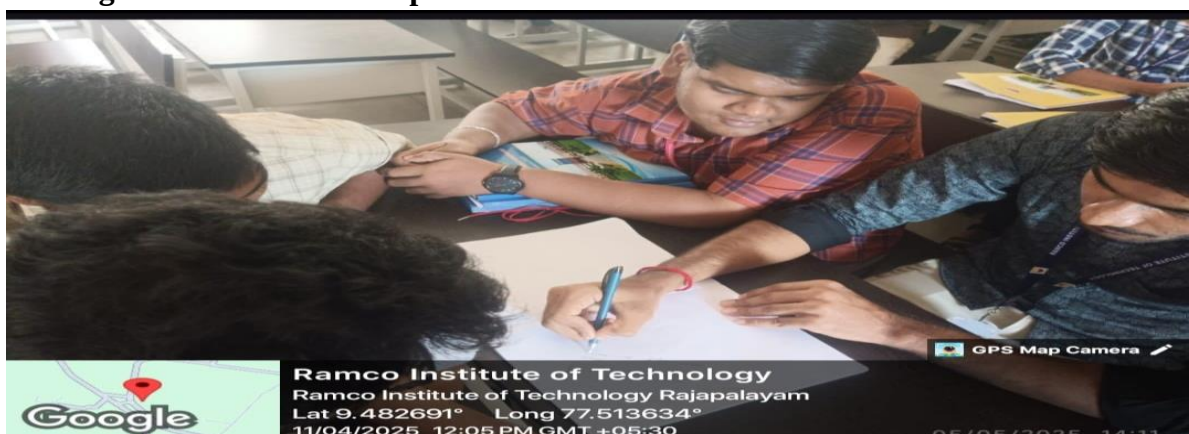


Fig 1: Students Sharing their Thinking.



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Fig 2: One of the Teams Presenting their idea

## Reflective Critique:

### Feedback of Practice from Students and Other Stakeholders:

- Students expressed that the activity enhanced their understanding of complex deployment tools.
- Peer discussion made tool comparisons and practical applications easier to grasp.
- The hands-on discussion prepared students better for interviews and real-time project implementation.

### Benefit of the Practice:

- Improved conceptual clarity of deployment tools and their applications.
- Better engagement than passive learning methods.
- Enabled students to apply knowledge in assignments, projects, and internships.
- Encouraged team learning and communication skills development.

### Challenges Faced in Implementation:

- Some students had no prior exposure to tools like Kubernetes, requiring additional explanation.
- A few pairs needed more time to complete the worksheet discussion and sharing.

### References:

1. <https://www.jenkins.io/doc/>
2. <https://www.docker.com/resources/what-container/>
3. <https://kubernetes.io/docs/concepts/overview/what-is-kubernetes/>
4. <https://www.geeksforgeeks.org/devops-tools-and-their-uses/>

Signature of Faculty Member

HOD